EE 206

Assignment

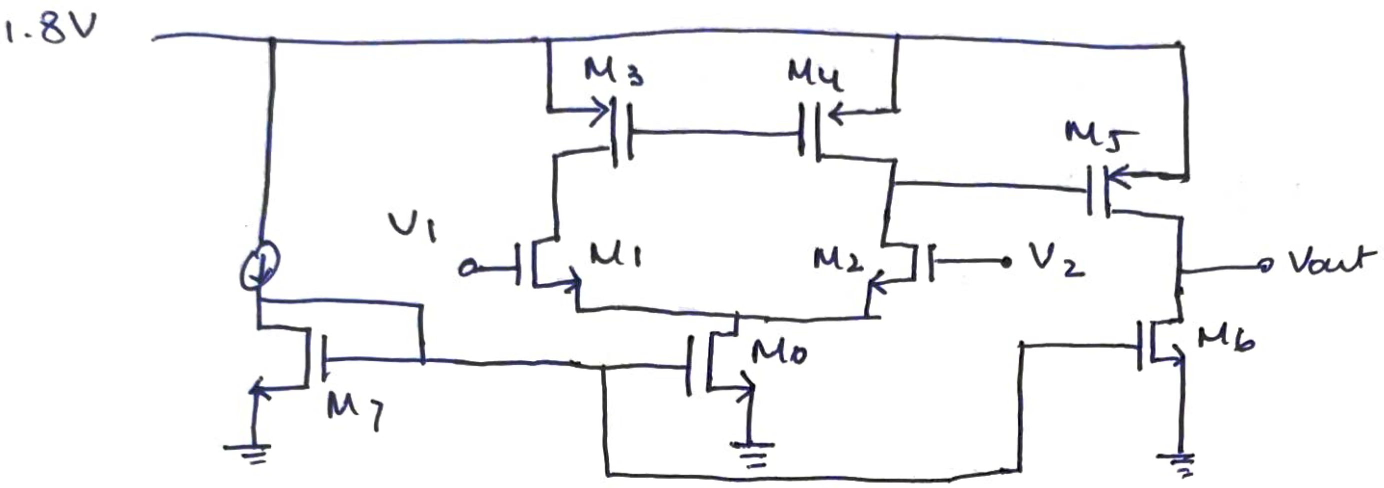
- Priyanshu Nayak
- 240102074.

Given: $V_{Tn} = 0.37V$, $V_{Tp} = 0.39V$

$\mu_n C_{ox} = 230 \mu A/V^2$, $\mu_p C_{ox} = 100 \mu A/V^2$

$V_{dd} = 1.8V$, $L_{min} = 0.18 \mu m$, $W_{min} = 0.27 \mu m$

Open loop circuit diagram.



Calculations:→

V_2 = AC signal of 0.1mV amplitude
V_1 = 0.9 Vdc
} for maxm swing.

for M_7 transistor:

- Assuming 0.2V overdrive, we get:→

  Iref chosen as 5μA:

  $$\Rightarrow I_{ref} = \frac{\mu_n C_{ox}}{2} \left(\frac{W}{L}\right)_7 (V_{gs} - V_{Tn})^2$$

  Solving we get,

  $$\left(\frac{W}{L}\right)_7 = 1.09.$$

Given: $L_{min} = 0.18 \mu m = 18 \times 10^{-8} = 180 nm$

$W_{min} = 0.27 \mu m = 270 nm$

$\therefore$ $L = 900 nm$ (lets take)

$W = \underline{98.1 nm}$

• for M_0: →

⇒ $I_0 = 10 I_{ref} = 50 \mu A$

As overdrive and $\mu_n C_{ox}$ are same

$$\Rightarrow \frac{(W/L)_0}{(W/L)_7} = 10$$

$$\therefore \left(\frac{W}{L}\right)_0 = 10.9$$

So, $L = 900nm$
$W = 9810nm$
$= 9.81 \mu m$

Also, $V_S > (V_{gs} - V_{th})$

$$\Rightarrow \boxed{V_S > 0.2V}$$

• for M_1 and M_2 transistor : →

I_1 and $I_2 = \frac{50}{2} \mu A = 25 \mu A$

choosing the overdrive $\underline{0.2V}$.

$$\frac{(W/L)_1}{(W/L)_0} = \frac{1}{2} \qquad \text{Cas } I_1 = I_2 = I_0/2$$

$$\Rightarrow \left(\frac{W}{L}\right)_1 = \frac{1}{2} \times 10.8 = 5.45$$

So, is $L_1 = L_2 = 900nm$.
$W_1 = W_2 = 4.91 \mu m$.

Now,

$$V_{GS} - V_{TH} = 0.2V$$

$$\Rightarrow 0.9V - V_S - 0.37V = 0.2V$$

$$\Rightarrow V_S = (0.9 - 0.37 - 0.2)V$$

$\therefore V_S = 0.33V > 0.2V$ so, M_0 is in saturation.

Now, $V_{D_1} > (V_{gs} - V_{th})_1$

$$\underline{\underline{V_{D_1} > 0.2V}}.$$

⇒ for M_3, M_4

$I_3 = I_4 = 25 \mu A$.

$$I_3 = \frac{1}{2} \mu_p C_{ox} (|V_{GS}| - |V_{TH}|)_3^2 \cdot \left(\frac{W}{L}\right)_3$$

$$\Rightarrow \frac{25}{50} = \left(\frac{W}{L}\right)_3 (|V_{GS}| - |V_{TH}|)_3^2$$

Taking $\left(\frac{W}{L}\right)_3 = 10$

$$\boxed{(|V_{GS}| - |V_{TH}|)_3 = 0.223V}$$

So, $(V_{SG} - V_{th})_4 = 0.223V$

and $w_3 = w_4 = 9\mu m$

$L_3 = L_4 = 900nm$

Now, for M_3 to be ON,

$V_{SG} > |V_{TP}|$

$\Rightarrow V_{SG} > |V_{TP}|$

$\Rightarrow 1.5V - |V_{TP}| > V_g$

$\Rightarrow V_g < (1.8 - 0.39)V$

$\therefore V_g < \underline{1.91V}$

We get, $(V_{SG} - V_{Th})_3 = 0.223V$

$\Rightarrow 1.8V = 0.223 + V_{D_1} + 0.33$

$$\boxed{V_{D_1} = 1.189V}$$

Clearly it satisfies $V_{D_1} > 0.2V$ and $V_{D_1} < 1.41V$

So M_3 and M_4, M_1 and M_2 } Are in saturation.

$\therefore$ Gain due to first stage $= g_{m_2}(r_{oN_2} \| r_{oP_4})$

(taken from logfile) $= -102.06$

practically $= -83.02$

$\therefore$ $\boxed{\text{error} = 18\%}$

practically, $V_{D_1} = V_{mid} \approx 1.102V$ (as Vin varies due to body effect)

* $\underline{M_6}$: $I_b = 25\mu A$ as

$V_{g7} = V_{g6}$

$\Rightarrow 25\mu A = \frac{1}{2}(230\mu \times \left(\frac{W}{L}\right)_6 \times (0.2)^2$

$\Rightarrow \left(\frac{W}{L}\right)_6 = \frac{25 \times 2}{230 \times 0.2^2}$

$\therefore \left(\frac{W}{L}\right)_6 = 5.435$

$\therefore L_{6C} = 900nm$

$W_6 = 4.892 \mu m$

Practically $W_6 = 4.628 \mu m$

$L_6 = 900\ nm$

$I_6 = 29.3 \mu A$

* M_5: $(V_{GS} - V_{TN})_5 = V_S - V_{MW} - V_{TN}$

$= 1.8 - 1.187 - 0.39$

$= 0.223V$

$I_5 = 25 \mu A$

$\Rightarrow \left(\frac{W}{L}\right)_5 = \frac{25 \times 2}{100 \times (0.223)^2} = 10$

$L_5 = 900nm$

$W_5 = 9 \mu m$

So gain for second stage $= g_{m5} \times (r_{op5} \| r_{on6})$

$= -9.14 \times 10^{-5} \times (279329.608)$

$= -25.530$

$\therefore$ Theoretical gain $= 2605.59$

& practically, open loop gain of OTA :-

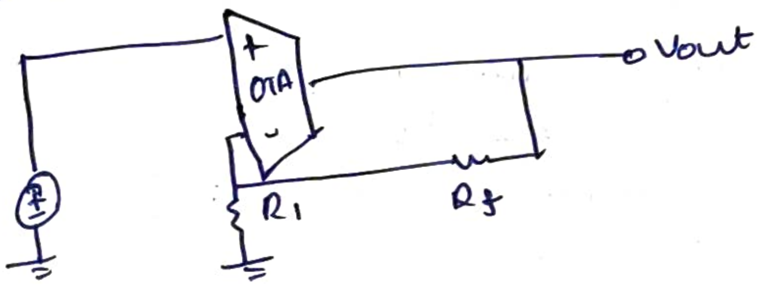
3027.82

first stage as 83.02

second stage as 36.47

$\therefore$ error $\simeq 16\%$

Closed loop:-



$V_{out} = (V_+ - V_-)A.$

$$\Rightarrow V_- = V_{in} - \frac{V_{out}}{A} = \frac{(AV_{in} - V_{out})}{A}$$

So KCL:

$$\frac{\frac{AV_{in} - V_{out}}{A}}{R_1} + \frac{\frac{(AV_{in} - V_{out})}{A} - V_{out}}{R_f} = 0$$

$$\Rightarrow V_{in}\left[\frac{R_1 + R_f}{R_1 R_f}\right] = V_{out}\left(\frac{R_f + AR_1 + R_1}{R_1 R_f}\right)$$

$$\Rightarrow \frac{V_{out}}{V_{in}} = \frac{A(R_1 + R_f)}{(1+A)R_1 + R_f}$$

$$= \left(1 + \frac{R_f}{R_1}\right)\left(\frac{A}{(1+A) + \frac{R_f}{R_1}}\right)$$

So choosing

$R_f = 21k\Omega$

$R_1 = 20k\Omega$

$A = 23$

$$\frac{V_{out}}{V_{in}} = \left(1 + \frac{21}{20}\right)\left(\frac{3027.82}{3027.82 + \frac{21}{20}}\right)$$

$$\therefore \frac{V_{out}}{V_{in}} \simeq 2.049$$

Also practically getting

$$\frac{V_{out}}{V_{in}} \simeq 2.035$$

$\therefore$ error $= -0.6\%$.

$C_L = 10pF$

$C_c = 3pF$

Power distribution $=$

$$(5 + 50 + 25) \times 1.8$$

$$= 1.44 \times 10^{-4}\, W$$

$$= 0.144\, \omega$$